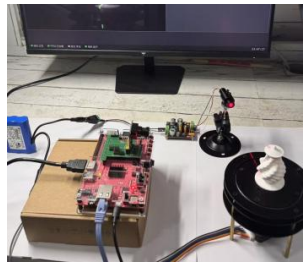
Intelligent Line-Laser 3D Measurement System

Embedded 3D measurement
Hardware / software co-design

AMD

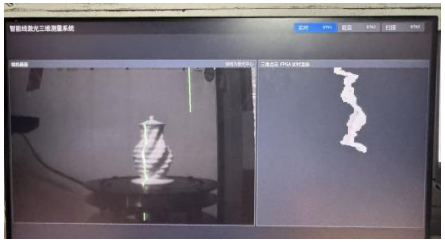


On-board test: PYNQ-Z2

INTRODUCTION

Single-board profile and dimension inspection, from camera capture to 3D reconstruction and display.

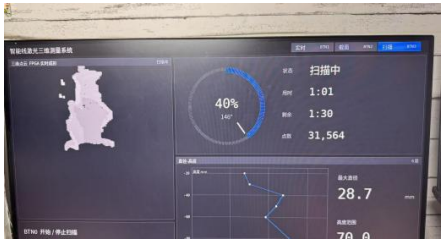
BTN1 · Live — Camera stripe and live point cloud



BTN2 · Section — Profile, measurement status and trend



BTN3 · Scan — Angle progress and accumulated cloud



Pixel-to-3D hardware pipeline

Data → Control / status

MT9V034 Camera
752 × 480 grayscale

PL / Programmable Logic — 100 MHz

- **Camera Capture / CDC FIFO**
  Frame identity and pixel stream
- **Shared Window / Hessian**
  Shared filter and derivative features
- **Direction Lane 0**
  Direction / refinement
- **Direction Lane 1**
  Direction / refinement
- **Stripe Extraction / Branch Linking**
  Q12 centres and quality flags
- **Fixed-point Geometry Engine**
  Undistort · triangulate · rotate
- **Measure / Section**
  AXI-Lite result registers
- **Voxel Renderer**
  Point buffer / projection
- **DMA / DDR3**
  Raw · PFO3 · XYZ packets
- **Video / HDMI Output**
  VDMA + PL cloud overlay

GP0 — Result readback

PS / ARM Cortex-A9
Scan control, stitching and assessment
HDMI pages, storage and export

Motion Guard / Turntable BTN0 scan · BTN1–3 pages

CREATIVE DESIGN

RESULTS

PL image processing

26.5 frames/s

Frame-by-frame processing, zero packet errors

Median stripe-centre error **0.02 px**

Synthetic stripes with known centre positions

Measurement	Result
360° scan	200 views / ≈2.5 min
Measured cloud	≈66,000 points
Stepped cylinder	−0.32 to −0.10 mm
Vase neck	−0.27 to +0.05 mm

Signed diameter differences relative to caliper readings.
Stepped-cylinder results are calibration rechecks.

Implemented FPGA resources

LUT 73.5% FF 47.8% BRAM 78.9% DSP 60.9%

WNS +0.230 ns On-chip power ≈2.37 W



Measured cylinder cloud Vase reconstructed surface

AMD